

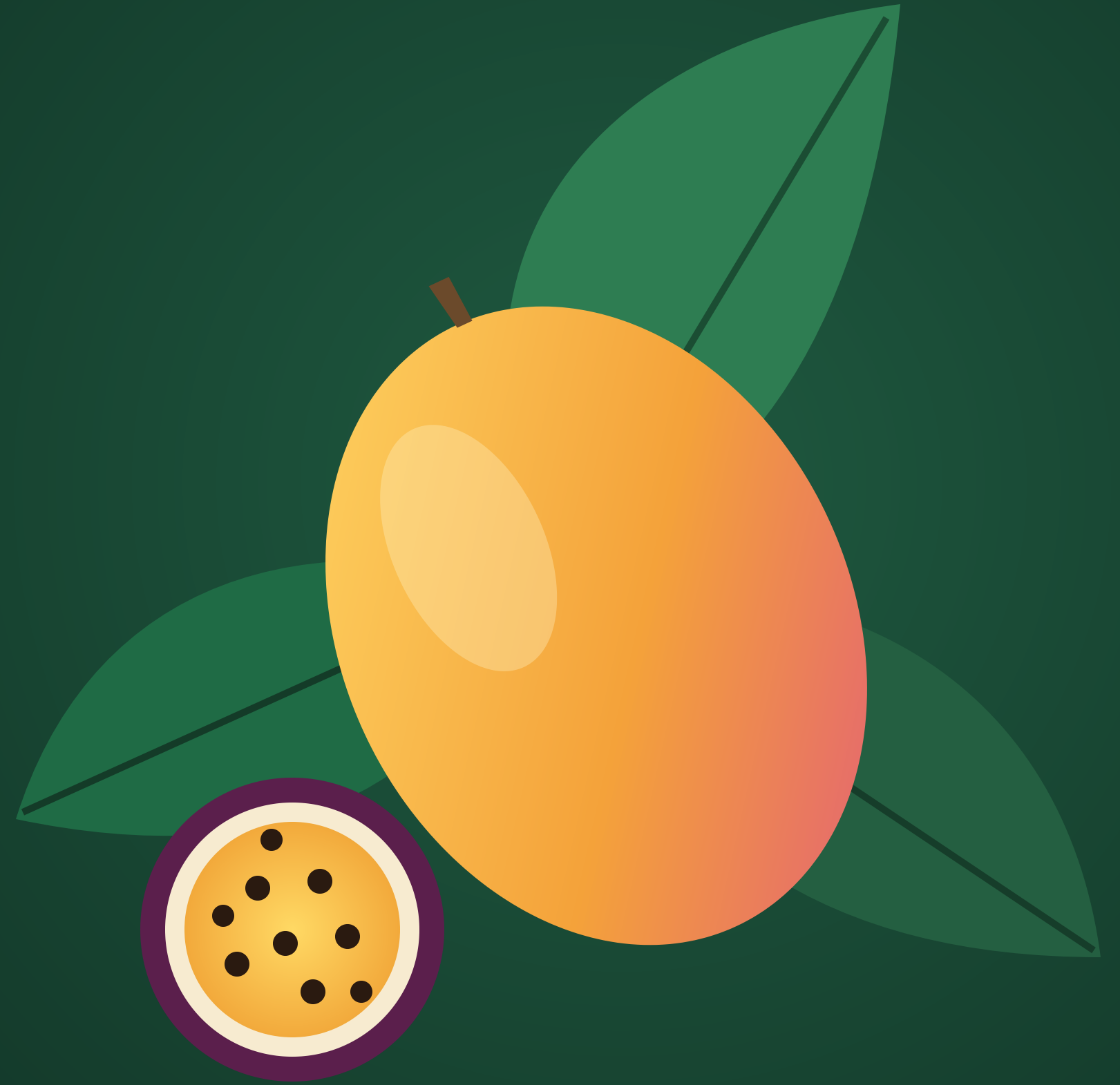
TERM 2 · PRODUCT IDENTIFICATION

Tropical *Fruits*

Category: **Fruits** · Sub-category: **Tropical & exotic fruits**

[Your Name] · [Class / Section] · [Date]

Based on *The Professional Chef*, 9th Edition — The Culinary Institute of America



INTRODUCTION

What is product identification?

A chef's ability to recognise an ingredient — its variety, quality, ripeness and best use — before it ever reaches the cutting board.

01

Identify

Name the product, its variety and its market forms.

02

Select

Judge quality, ripeness, seasonality and grade.

03

Use

Store, prepare and cook it to its full potential.

CHOSEN CATEGORY & SUB-CATEGORY

From the textbook to our topic



Source: The Culinary Institute of America, *The Professional Chef*, 9th Ed. (pp. 40–235 range)

CHOSEN TOPIC · INTRODUCTION

What makes a fruit *tropical*?

Fruits native to the hot, humid belt around the equator — sweet, fragrant, vividly coloured and loved worldwide.



Grown between the Tropics of Cancer and Capricorn (23.4° N to 23.4° S)



Frost-sensitive — need warmth all year round



Many ripen after picking, so they travel well when harvested firm



Once rare luxuries, now available year-round worldwide

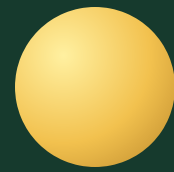
MEET THE FAMILY

Eight tropical stars



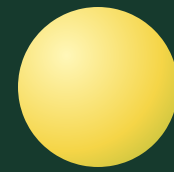
Mango

"King of fruits", native to South Asia



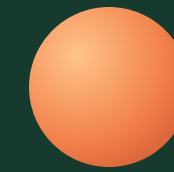
Pineapple

A bromeliad from South America



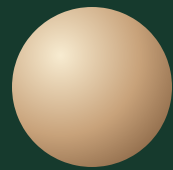
Banana

Grows on a giant herb, not a tree



Papaya

Central American, rich in papain



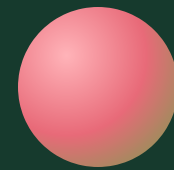
Coconut

Gives water, meat, milk and oil



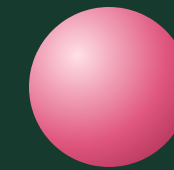
Passion fruit

Tart, perfumed pulp from Brazil



Guava

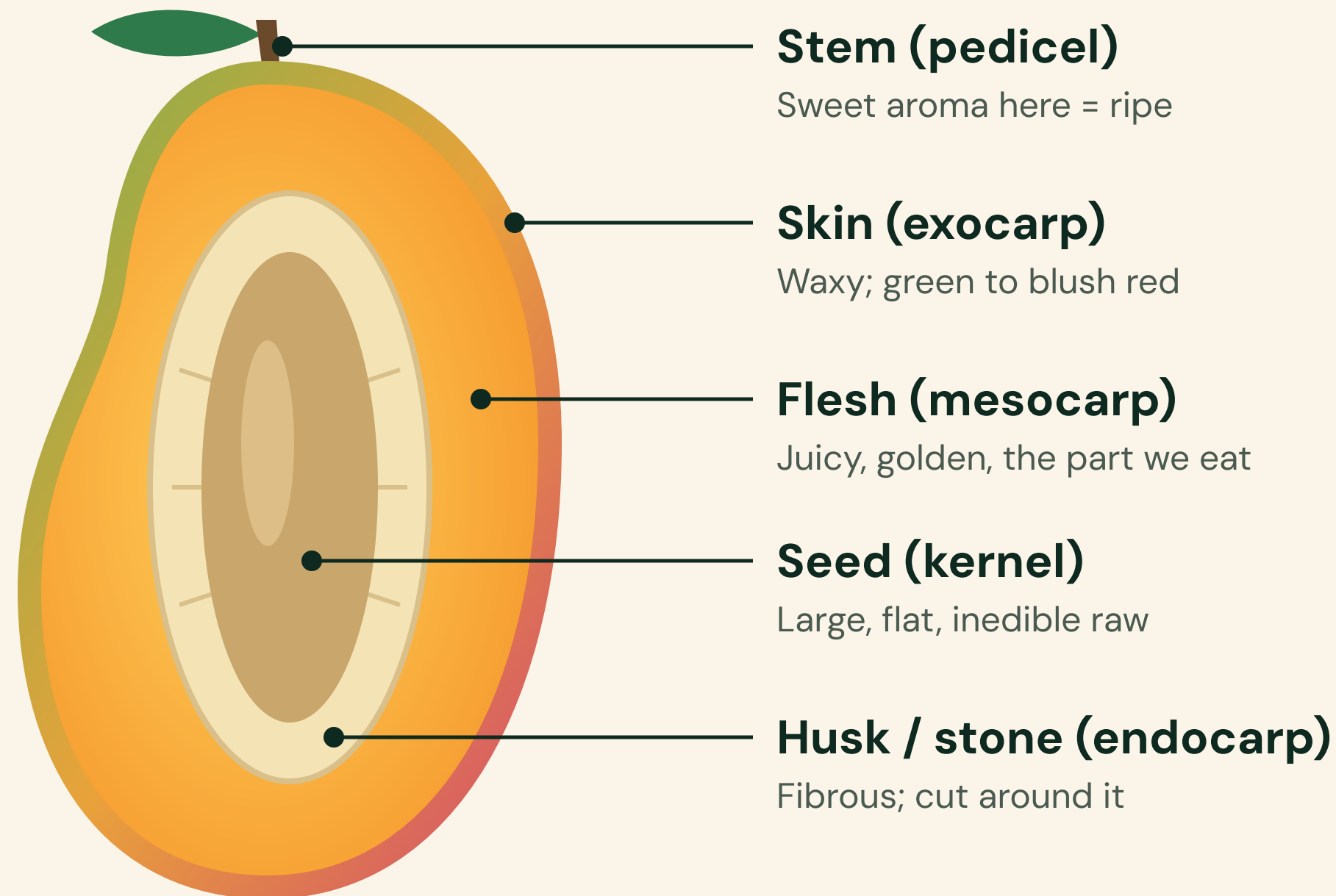
A vitamin C powerhouse



Lychee

Floral and juicy, from south China

Inside a mango



Characteristics

- A drupe (stone fruit) of the cashew family
- Climacteric: ripens after harvest
- Ripe = slight give and a fruity scent
- 1,000+ varieties: Chaunsa, Sindhri, Anwar Ratol, Alphonso, Ataulfo

How the others are built

Fruit	Botanical type	Key parts	Chef's note
Pineapple	Multiple fruit	Crown · eyes · flesh · core	Cut out the woody core
Banana	Berry	Peel · flesh · tiny seeds	Browns fast: add lemon
Papaya	Berry	Skin · flesh · seed cavity	Seeds taste peppery
Coconut	Fibrous drupe	Husk · shell · meat · water	Pierce the “eyes” first
Passion fruit	Berry	Rind · pith · seeded pulp	Wrinkled = ripest
Lychee	Drupe-like	Shell · aril (flesh) · seed	Peel like an egg

A pineapple is 100–200 small flowers that fuse into one fruit around a central core.

Choosing a perfect fruit

Fruit	Look for	Avoid
Mango	Fruity scent at the stem, slight give	Shrivelled skin, sunken black spots
Pineapple	Heavy for size, sweet smell at the base	Fermented smell, soft or wet spots
Banana	Even yellow with a few brown freckles	Dull grey skin (chilled), splits
Papaya	Yellow-orange skin, gives to light press	Mould, very soft patches
Coconut	Heavy, water sloshes when shaken	Damp or mouldy “eyes”, cracks

Climacteric — keep ripening after picking

Mango · banana · papaya · guava · passion fruit

Non-climacteric — buy them ripe

Pineapple · lychee · dragon fruit

ETYMOLOGY

Where the names come from

Mango

Tamil *mānkāy* → Malay *mangga* → Portuguese *manga*

Pineapple

English for its pine-cone look; *ananas* from Tupi *nanas*

Banana

A West African word, carried by Portuguese and Spanish traders

Papaya

From a Caribbean (Carib/Arawak) word, via Spanish

Coconut

Portuguese *coco*, “grinning face”: the three eyes look like one

Guava

Spanish *guayaba*, from the Taíno language of the Caribbean

AGRICULTURE

What the trees need

20–30 °C

Warm days all year for steady growth

No frost

Even one cold night can kill young plants

Rain + sun

High rainfall or irrigation, well-drained soil

Mango — India and Pakistan (Chaunsa, Sindhri, Anwar Ratol)

Banana — India grows most; Ecuador exports most

Pineapple — Costa Rica, Philippines, Indonesia

Coconut — Indonesia, Philippines, India, Sri Lanka

BREEDING & CULTIVATION

How new plants are made

Mango

Grafted onto seedling rootstock so each variety stays true to type

Pineapple

Grown from crowns, slips and suckers; flowering is timed with ethylene

Banana

Seedless clones, spread by suckers and tissue culture

Papaya

Grown from seed; plants are male, female or both



The banana lesson

The Gros Michel banana was wiped out by Panama disease in the 1950s. Today's Cavendish clone faces a new strain, TR4 — why breeders seek resistant varieties.

PRODUCTION

From farm to kitchen



Chilling injury: most tropical fruit is damaged below about 10–13 °C — bananas turn grey and mangoes fail to ripen. Never fridge them unripe.

Beyond the fresh fruit

Canned

Pineapple rings and chunks in juice or syrup; lychees; mango slices

Pulp & purée

Mango pulp, passion fruit purée — staples of the pastry kitchen

Dried & candied

Dried mango and papaya, banana chips, candied pineapple

Coconut products

Milk, cream, oil, water, desiccated flakes, coconut sugar

Juice & concentrate

Pasteurised juices, nectars and frozen concentrates

Frozen (IQF)

Individually quick-frozen chunks for smoothies and sorbets

NUTRITION • PER 100 G, RAW

Sweet, and good for you

Fruit	kcal	Carbs g	Fibre g	Vit C mg	Stand-out
Mango	60	15.0	1.6	36	Vitamin A (beta-carotene)
Pineapple	50	13.1	1.4	48	Manganese, bromelain
Banana	89	22.8	2.6	9	Potassium (358 mg)
Papaya	43	10.8	1.7	61	Folate, papain
Guava	68	14.3	5.4	228	4× the vitamin C of an orange
Passion fruit	97	23.4	10.4	30	Very high fibre (seeds)
Coconut meat	354	15.2	9.0	3	33 g fat, mostly saturated

Values approximate. Source: USDA FoodData Central.

Fresh pineapple *eats* protein.

Pineapple (bromelain), papaya (papain) and kiwi (actinidin) contain enzymes that break down proteins.

Tenderises meat

Great in marinades — but hours too long turns meat mushy

Stops gelatin setting

Cook the fruit first, or use canned — heat kills the enzyme

Turns dairy bitter

Fresh pineapple in yogurt or cream breaks it down if left to stand

TOXICITY & ALLERGIES

Handle with knowledge

Mango skin

Contains urushiol, as in poison ivy; can cause a rash around the mouth

Latex-fruit allergy

People allergic to latex may react to banana, papaya, kiwi, avocado

Akee

Unripe fruit and seeds hold hypoglycin A — eat only ripe, opened arils

Star fruit

Oxalates and a neurotoxin: dangerous for people with kidney disease

Unripe papaya

Its latex is traditionally avoided during pregnancy

Wash before cutting

A knife carries surface bacteria from the rind into the flesh

Prep like a professional

Mango “hedgehog”

1. Stand it on its end; slice each cheek off beside the flat stone
2. Score the flesh in a grid without cutting the skin
3. Push the skin up so the cubes pop out
4. Trim the flesh left around the stone

Peeling a pineapple

1. Cut off the crown and base so it stands flat
2. Slice the skin away top to bottom, following the curve
3. Remove the eyes in diagonal V-shaped grooves
4. Quarter lengthwise and cut out the core

Storage rules

- Ripen at room temperature, away from sun
- Paper bag + a banana speeds ripening
- Refrigerate only once ripe, for 2–3 days
- Cut fruit: covered, chilled, used within a day

CULTURAL SIGNIFICANCE

More than food

Mango, national fruit

Of Pakistan, India and the Philippines; Pakistan sends mangoes abroad as diplomatic gifts

Pineapple welcome

A symbol of hospitality and luxury in Europe and colonial America

Coconut, “tree of life”

Called *kalpavriksha* in South Asia; broken at Hindu ceremonies for good luck

Banana leaf plates

South Indian feasts are served on the leaf; it also wraps food for steaming

The emperor’s lychee

Tang China ran horse relays to bring fresh lychees to consort Yang Guifei

Mango season

Summer mango festivals across South Asia celebrate local varieties

CULINARY USES

Sweet, savoury and everything between

Application	Examples
Fresh & raw	Fruit platters, salads, salsas, chaat with chaat masala
Savoury	Som tum, mango chutney, pineapple fried rice, tacos al pastor
Baking & pastry	Banana bread, pineapple upside-down cake, coconut macaroons
Frozen desserts	Mango kulfi, passion fruit sorbet, coconut ice cream
Beverages	Mango lassi, piña colada, aguas frescas, fresh coconut water
Preserves	Raw mango achar, guava jelly, papaya jam, candied peel

Mango Lassi

Serves 2

Prep 5 min

No cooking



Ingredients

- 2 ripe mangoes (Sindhri or Chaunsa), about 300 g flesh
- 250 g plain yogurt, chilled
- 120 ml cold milk
- 2–3 tbsp sugar or honey
- 2 green cardamom pods, seeds crushed
- Ice, chopped pistachios, saffron

Method

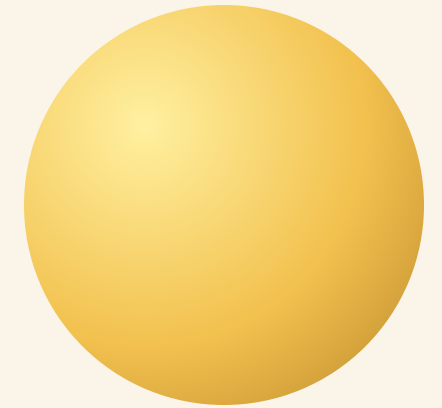
1. Peel the mangoes and cut the flesh away from the stone.
2. Blend mango, yogurt, milk, sugar and cardamom until silky, about 1 minute.
3. Taste; add sugar if needed, or a splash of milk if too thick.
4. Pour over ice in chilled glasses.
5. Top with pistachios and a few saffron threads. Serve at once.

Grilled Pineapple, Chili-Lime Honey

Serves 4

Prep 15 min

Cook 8 min



Ingredients

- 1 ripe pineapple
- 3 tbsp honey
- Zest and juice of 1 lime
- ½ tsp chili flakes
- Pinch of flaky salt
- Fresh mint leaves
- Vanilla ice cream, to serve

Method

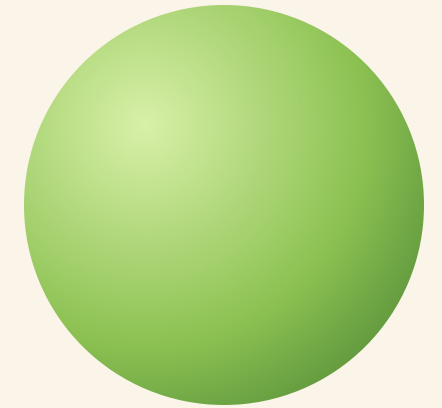
1. Peel, remove the eyes and cut into 2 cm spears; cut out the core.
2. Warm honey, lime juice, zest and chili until loose.
3. Grill on high heat, 3–4 minutes a side, for deep char marks.
4. Brush with the chili-lime honey in the last minute.
5. Finish with salt and torn mint; serve warm with ice cream.

Som Tum, Green Papaya Salad

Serves 2

Prep 20 min

No cooking



Ingredients

- 300 g green papaya, julienned
- 1–2 bird's-eye chilies, 2 garlic cloves
- 1 tbsp palm sugar
- 2 tbsp fish sauce, 2 tbsp lime juice
- 6 cherry tomatoes, halved
- 50 g green beans, in 3 cm lengths
- 2 tbsp roasted peanuts

Method

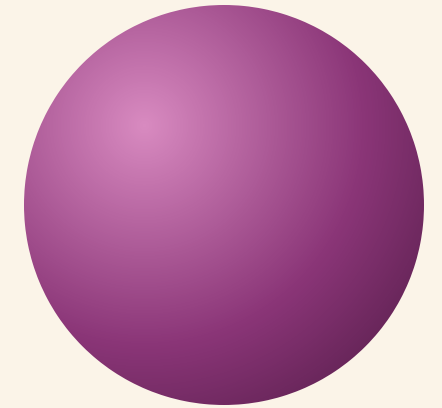
1. Peel and seed the papaya; shred finely and soak in iced water.
2. Pound garlic and chilies in a mortar to a rough paste.
3. Add sugar, fish sauce and lime; pound until dissolved.
4. Bruise beans and tomatoes, then add papaya, pounding and tossing.
5. Balance sweet, sour, salty, hot; top with peanuts and serve.

Passion Fruit Panna Cotta

Serves 6

Prep 20 min

Set 4 hours



Ingredients

- 500 ml double cream
- 120 ml whole milk
- 80 g sugar, plus 1 tbsp for the topping
- 1 vanilla pod, split
- 4 gelatin sheets (8 g)
- 6 passion fruits (about 120 g pulp)

Method

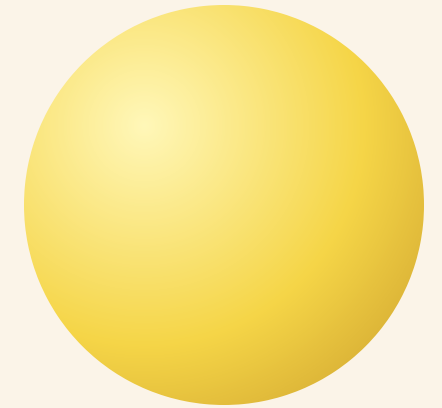
1. Soak the gelatin in cold water for 5 minutes.
2. Bring cream, milk, sugar and vanilla to a simmer; take off the heat.
3. Squeeze the gelatin, whisk it into the hot cream, then strain.
4. Pour into 6 moulds; chill 4 hours until set with a gentle wobble.
5. Warm the pulp with 1 tbsp sugar, cool, and spoon over to serve.

Coconut Banana Fritters

Serves 4

Prep 15 min

Fry 10 min



Ingredients

- 4 ripe but firm bananas, halved
- 100 g rice flour, 30 g plain flour
- ½ tsp baking powder
- About 150 ml coconut milk
- 2 tbsp desiccated coconut
- 1 tbsp sugar, pinch of salt
- Oil for deep-frying

Method

1. Whisk the flours, baking powder, sugar, salt and coconut.
2. Add coconut milk slowly to make a thick, smooth batter.
3. Heat the oil to 180 °C in a deep, heavy pan.
4. Dip the bananas in batter; fry in batches 3–4 minutes until golden.
5. Drain on a rack; serve hot with palm-sugar syrup or icing sugar.

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Thank you — shukriya!